

Unifying Special Relativity and Quantum Mechanics

Lecture 3 — The Poincaré Algebra in Covariant Form; Unitary Irreps of the Poincaré Group: the Massive Case and Wigner’s Little Group

(Typed version of the handwritten notes, with added explanations in blue boxes)

Added explanation (not in the original notes)

How to read this document. Same conventions as Lectures 1–2: normal text and **yellow boxes** are a faithful transcription of the handwritten notes; **blue boxes** are my additions. Two remarks on the pagination of this scan: (i) the first two scanned pages are pp. 43–44, which *repeat* the last two pages of Lecture 2 (boost examples and the start of the tensor notation) — they were already typed there, so this document starts from the point where p. 44 left off, i.e. the step marked $\circledast$; (ii) after p. 51 the notes restart their page numbering from 1 for the new chapter “Unitary irreps of the Poincaré group”. Markers for that part are written [p. 1*], ..., [p. 8*] to avoid confusion with the first lecture.

Where we left off. Lecture 2 introduced the six Lorentz generators $\vec{J}$ (rotations) and $\vec{K}$ (boosts), their commutators, and began packing them into an antisymmetric tensor $J^{\mu\nu}$ with parameters $\omega_{\mu\nu}$, defining $\theta^k \equiv \frac{1}{2}\varepsilon_{kij}\omega^{ij}$, $J^k \equiv \frac{1}{2}\varepsilon_{kij}J^{ij}$, $\omega^{i0} = \omega_{0i} \equiv \eta^i$, $J^{0i} \equiv K^i$. The first computation below closes that circle.

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1 The Lorentz algebra in covariant form [p. 45]

Continuing from the chain of equalities started on p. 44:

$$\stackrel{\circledast}{=} \exp\left(-\frac{i}{2}\varepsilon_{ijk}\theta^k J^{ij} - i\eta^k K^k\right) = \exp\left(-i\theta^k J^k - i\eta^k K^k\right) = \exp\left(-i\vec{\theta} \cdot \vec{J} - i\vec{\eta} \cdot \vec{K}\right),$$

using the “inverse relation” $\omega^{ij} = \varepsilon_{ijk}\theta^k$. We thus have, in covariant form:

$$D[\Lambda(\omega)] = \exp\left(-\frac{i}{2}\omega_{\mu\nu}J^{\mu\nu}\right) \equiv D(\omega)$$

$\hookrightarrow \omega$ is a 4×4 antisymmetric matrix that contains all the 6 parameters needed to identify a Lorentz transformation.

In the four-vector representation:

$$\Lambda(\omega) = \exp \left(-\frac{i}{2} \omega_{\mu\nu} J_{4\text{vec}}^{\mu\nu} \right).$$

Added explanation (not in the original notes)

Quick check of the “inverse relation”: from $\theta^k = \frac{1}{2} \varepsilon_{kij} \omega^{ij}$, contract both sides with ε_{kmn} and use $\varepsilon_{kmn} \varepsilon_{kij} = \delta_{mi} \delta_{nj} - \delta_{mj} \delta_{ni}$: one finds $\varepsilon_{kmn} \theta^k = \frac{1}{2} (\omega^{mn} - \omega^{nm}) = \omega^{mn}$. The same relation holds between J^k and J^{ij} , which is why $\frac{1}{2} \omega_{ij} J^{ij} = \theta^k J^k$: each antisymmetric pair (ij) is counted twice in the sum and the $\frac{1}{2}$ removes the double counting.

Infinitesimal transformations. Consider the case of infinitesimal transformations: take ω as infinitesimal parameters. In components, we write

$$\Lambda(\omega)^\mu{}_\nu = \exp \left(-\frac{i}{2} \omega_{\rho\sigma} J_{4\text{vec}}^{\rho\sigma} \right)^\mu{}_\nu = \left[\mathbb{K} - \frac{i}{2} \omega_{\rho\sigma} J_{4\text{vec}}^{\rho\sigma} \right]^\mu{}_\nu = \delta^\mu{}_\nu - \frac{i}{2} \omega_{\rho\sigma} (J_{4\text{vec}}^{\rho\sigma})^\mu{}_\nu \stackrel{\textcircled{*}}{=}$$

[p. 46] For fixed (ρ, σ) , $(J_{4\text{vec}}^{\rho\sigma})^\mu{}_\nu$ is a four-dimensional matrix with components (μ, ν) . We constructed these matrices explicitly before. There is actually a nice simple formula for these objects. We get

$$(J_{4\text{vec}}^{\rho\sigma})^\mu{}_\nu = i(g^{\mu\rho} \delta^\sigma{}_\nu - g^{\mu\sigma} \delta^\rho{}_\nu).$$

$$\stackrel{\textcircled{*}}{=} \delta^\mu{}_\nu - \frac{i}{2} \omega_{\rho\sigma} i(g^{\mu\rho} \delta^\sigma{}_\nu - g^{\mu\sigma} \delta^\rho{}_\nu) = \delta^\mu{}_\nu + \omega_{\rho\sigma} \frac{2}{2} g^{\mu\rho} \delta^\sigma{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu$$

$$\Lambda(\omega)^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu.$$

Added explanation (not in the original notes)

Checking the master formula against the explicit generators. Take $J^3 = J^{12}$: the formula gives $(J^{12})^\mu{}_\nu = i(g^{\mu 1} \delta^\sigma{}_\nu - g^{\mu 2} \delta^\sigma{}_\nu)$, whose only nonzero entries are $(J^{12})^1{}_2 = ig^{11} = -i$ and $(J^{12})^2{}_1 = -ig^{22} = +i$ — exactly the matrix $J_{4\text{vec}}^3$ of Lecture 2 (p. 37). Similarly $K^3 = J^{03}$: nonzero entries $(J^{03})^0{}_3 = ig^{00} = +i$ and $(J^{03})^3{}_0 = -ig^{33} = +i$, matching $K_{4\text{vec}}^3$ (p. 38). ✓

In the middle step of the chain, the two terms of the formula give equal contributions after using the antisymmetry $\omega_{\rho\sigma} = -\omega_{\sigma\rho}$ (relabel $\rho \leftrightarrow \sigma$ in the second term), which cancels the overall $\frac{1}{2}$ — this is the cancellation the notes indicate with the struck-out 2's. The result $\Lambda = \mathbb{K} + \omega$ with the *first index up, second down* is consistent with the Lorentz condition: lowering the first index, $\omega_{\mu\nu}$ must be antisymmetric, which is exactly the statement $X^T g = -g X$ of p. 35 at first order.

2 The Lie algebra of space-time translations

[p. 46]

Space-time translations are non-linear. Therefore, to visualize their effect in matrix form, we use the embedding homomorphism. We write [p. 47]

$$\Phi(P(\Lambda, a)) = \left(\begin{array}{c|c} \Lambda & a^0 \\ \hline \vec{0}^T & 1 \end{array} \right) \in GL(5, \mathbb{R}) \quad (\Lambda: 4 \times 4 \text{ block}).$$

We focus on translations, we set $\Lambda = \mathbb{K}$:

$$\Phi(T(a)) = \left(\begin{array}{c|c} \mathbb{K} & \begin{matrix} a^0 \\ \vec{a} \end{matrix} \\ \hline \vec{0}^T & 1 \end{array} \right) \quad \hookrightarrow \quad \Phi(T(a)) \begin{pmatrix} x^0 \\ \vec{x} \\ 1 \end{pmatrix} = \begin{pmatrix} x^0 + a^0 \\ \vec{x} + \vec{a} \\ 1 \end{pmatrix}.$$

The Lie algebra is given by all matrices X such that

$$e^{tX} = \left(\begin{array}{c|c} \mathbb{K} & \begin{matrix} a^0 \\ \vec{a} \end{matrix} \\ \hline \vec{0}^T & 1 \end{array} \right), \quad \forall t \in \mathbb{R} \quad (\star)$$

If we consider a matrix

$$X = \left(\begin{array}{c|c} 0 & \begin{matrix} y^0 \\ \vec{y} \end{matrix} \\ \hline \vec{0}^T & 0 \end{array} \right),$$

we see immediately that $X^k = 0$ (for $k \geq 2$) and, consequently,

$$e^{tX} = \mathbb{K} + tX = \left(\begin{array}{c|c} \mathbb{K} & \begin{matrix} t y^0 \\ t \vec{y} \end{matrix} \\ \hline \vec{0}^T & 1 \end{array} \right)$$

which has the correct form to match eq. $(\star) \forall t \in \mathbb{R}$.

Added explanation (not in the original notes)

This is the specialization to the translation subgroup of the general $\mathfrak{se}(m)$ computation of Lecture 1 (pp. 11–12): there, with a nonzero rotational block Y , one had $X^k = \begin{pmatrix} Y^k & Y^{k-1}\vec{v} \\ \vec{0}^T & 0 \end{pmatrix}$; setting $Y = 0$ kills every power beyond the first — the matrix is *nilpotent* — and the exponential series truncates after two terms. That is the matrix incarnation of the obvious fact that composing translations just adds the translation vectors, with no curvature effects.

[p. 48] We thus find that a generic element of the form

$$X = \left(\begin{array}{c|c} 0 & \begin{matrix} y^0 \\ \vec{y} \end{matrix} \\ \hline \vec{0}^T & 0 \end{array} \right) \quad \text{belongs to the Lie algebra.}$$

$\leadsto$ Consequently, a basis is given by the matrices

$$T^0 = \left(\begin{array}{c|c} 0 & \begin{matrix} 1 \\ \vec{0} \end{matrix} \\ \hline \vec{0} & 0 \end{array} \right), \quad T^1 = \left(\begin{array}{c|c} 0 & \begin{matrix} 0 \\ 1 \\ \vec{0} \end{matrix} \\ \hline \vec{0} & 0 \end{array} \right), \dots$$

where T^0 generates translations in time and T^1 generates translations along $\hat{x}$, so that a generic element of the algebra can be written as

$$X = a_0 T^0 + a^1 T^1 + \dots$$

$\hookrightarrow$ In the physicist convention, we adopt the following definition:

$$X = a_0 T^0 + a^k T^k = i a_0 \underbrace{(-iT^0)}_{\equiv P^0} - i a^k \underbrace{(iT^k)}_{\equiv P^k} = i a_0 P^0 - i a^k P^k = i a_\mu P^\mu,$$

so that the generators take the form [p. 49]

$$P_{4\text{vec}}^0 = \left(\begin{array}{c|c} 0 & \begin{matrix} -i \\ 0 \\ 0 \\ 0 \end{matrix} \\ \hline 0 & 0 \end{array} \right), \quad P_{4\text{vec}}^1 = \left(\begin{array}{c|c} 0 & \begin{matrix} 0 \\ i \\ 0 \\ 0 \end{matrix} \\ \hline 0 & 0 \end{array} \right), \dots$$

The commutators are

$$[P^\mu, P^\nu] = 0$$

and the exponential map reads

$$D(a^\mu) = \exp(i a_\mu P^\mu) = \exp(i a_0 P^0 - i \vec{a} \cdot \vec{P})$$

$\hookrightarrow$ generic space-time translation by a constant four-vector a^μ (valid in *all* representations).

Added explanation (not in the original notes)

Why the relative minus sign between P^0 and P^k . The definitions $P^0 \equiv -iT^0$, $P^k \equiv +iT^k$ are engineered so that the four objects P^μ assemble into a *four-vector* of operators and the exponent becomes the manifestly covariant contraction $i a_\mu P^\mu = i a^\mu P_\mu$ (the metric supplies the sign flip $a_k = -a^k$, absorbing the asymmetric definition). The spatial part reproduces exactly the $SE(2)$ convention of Lecture 1, $e^{-i\vec{a} \cdot \vec{P}}$; the time part $e^{+i a_0 P^0}$ is fixed relative to it by covariance. Note also that P^0 will be identified with the *Hamiltonian* (energy operator): translations in time are generated by the energy, translations in space by the momentum — and $[P^\mu, P^\nu] = 0$ contains, as its $0i$ component, the statement that momentum is conserved ($[H, \vec{P}] = 0$), i.e. the starting point of Lecture 1, p. 1, now recognized as part of the Poincaré algebra. (Different textbooks distribute these i 's and signs differently; only the commutation relations and the group law are convention-independent.)

3 Transformation properties of P^μ

[p. 49]

Consider the following chain of transformations, assuming a to be infinitesimal:

$$D(\Lambda(\omega))^{-1} D(a^\mu) D(\Lambda(\omega)).$$

We compute this quantity in two ways as follows.

i) Using the composition rule:

$$D(\Lambda(\omega))^{-1} D(a) D(\Lambda(\omega)) = D(\Lambda(\omega)^{-1}) D(\Lambda(\omega), a) = D(\Lambda(\omega)^{-1} a)$$

[p. 50] ($\hookrightarrow$ still a pure space-time translation)

$$= \mathbb{I} + i [\Lambda(\omega)^{-1} a]_\rho P^\rho = \mathbb{I} + i (\Lambda(\omega)^{-1})^\mu_\rho a_\mu P^\rho = \mathbb{I} + i \Lambda(\omega)^\mu_\rho a_\mu P^\rho.$$

ii) By directly expanding:

$$D(\Lambda(\omega))^{-1} (\mathbb{I} + i a_\mu P^\mu) D(\Lambda(\omega)) = \mathbb{I} + i a_\mu D(\Lambda(\omega))^{-1} P^\mu D(\Lambda(\omega)).$$

From the comparison, we thus get

$$D(\Lambda)^{-1} P^\mu D(\Lambda) = \Lambda^\mu{}_\rho P^\rho.$$

Added explanation (not in the original notes)

Same two-pronged strategy as for $SE(2)$ (Lecture 1, pp. 20–21): sandwich an infinitesimal translation between $D(\Lambda)^{-1}$ and $D(\Lambda)$, evaluate it once with the group composition rule (Poincaré version: $P(\Lambda, 0)^{-1} P(\mathbb{K}, a) P(\Lambda, 0) = P(\mathbb{K}, \Lambda^{-1}a)$ — the translation subgroup is *normal*) and once by expanding, then match coefficients of the arbitrary a_μ . The middle step in (i) uses the index identity $(\Lambda^{-1})^\mu{}_\rho = \Lambda^\mu{}_\rho$ from Lecture 2 (p. 32). The boxed result says the four operators P^μ transform among themselves *as the components of a four-vector* — the operator counterpart of $x'^\mu = \Lambda^\mu{}_\nu x^\nu$.

As a further step, we can also take ω to be infinitesimal, meaning that, in the previous transformation law,

$$\underbrace{D(\Lambda(\omega))^{-1}}_{\mathbb{K} + \frac{i}{2} \omega_{\rho\sigma} J^{\rho\sigma}} P^\mu \underbrace{D(\Lambda(\omega))}_{\mathbb{K} - \frac{i}{2} \omega_{\rho\sigma} J^{\rho\sigma}} = \underbrace{\Lambda(\omega)^\mu{}_\rho}_{\delta^\mu{}_\rho + \omega^\mu{}_\rho} P^\rho.$$

[p. 51]

$$\begin{aligned} \left(\mathbb{K} + \frac{i}{2} \omega_{\rho\sigma} J^{\rho\sigma} \right) P^\mu \left(\mathbb{K} - \frac{i}{2} \omega_{\rho\sigma} J^{\rho\sigma} \right) &= (\delta^\mu{}_\rho + \omega^\mu{}_\rho) P^\rho \\ P^\mu + \frac{i}{2} \omega_{\rho\sigma} [J^{\rho\sigma}, P^\mu] &= P^\mu + \omega^\mu{}_\rho P^\rho, \end{aligned}$$

where

$$\omega^\mu{}_\rho P^\rho = g^{\mu\sigma} \omega_{\sigma\rho} P^\rho = -g^{\mu\sigma} \omega_{\rho\sigma} P^\rho = -\frac{1}{2} \omega_{\rho\sigma} (g^{\mu\sigma} P^\rho - g^{\mu\rho} P^\sigma).$$

From the comparison:

$$[J^{\rho\sigma}, P^\mu] = i(P^\rho g^{\mu\sigma} - P^\sigma g^{\mu\rho}).$$

In $3 + 1$ components, we extract

$$[J^k, P^0] = 0, \quad [J^k, P^i] = i \varepsilon_{kij} P^j.$$

Added explanation (not in the original notes)

Completing the $3 + 1$ decomposition (the boost components). The notes extract the rotation components; the same covariant bracket with $\rho = 0, \sigma = i$ (i.e. $J^{0i} = K^i$) gives the boost components:

$$[K^i, P^0] = -iP^i, \quad [K^i, P^j] = -i\delta^{ij}P^0.$$

Physical reading: rotations leave the energy P^0 alone and rotate $\vec{P}$ as a vector; boosts *mix* energy and momentum, exactly as a finite boost mixes E and $|\vec{p}|$ on the mass shell (Lecture 2, p. 43).

The full Poincaré algebra — now complete. Collecting everything obtained so far:

$$\begin{aligned} [P^\mu, P^\nu] &= 0, & [J^{\rho\sigma}, P^\mu] &= i(P^\rho g^{\mu\sigma} - P^\sigma g^{\mu\rho}), \\ [J^{\mu\nu}, J^{\rho\sigma}] &= i(g^{\nu\rho} J^{\mu\sigma} - g^{\mu\rho} J^{\nu\sigma} - g^{\nu\sigma} J^{\mu\rho} + g^{\mu\sigma} J^{\nu\rho}), \end{aligned}$$

the last line being the covariant packaging of the $[J, J]$, $[J, K]$, $[K, K]$ brackets of Lecture 2. These ten generators (4 momenta + 6 Lorentz) and their brackets are *the* input for everything that follows: the unitary irreps constructed next are representations of precisely this algebra.

4 Unitary irreps of the Poincaré group: massive case

[p. 1*]

4.1 Step 1: definition of momentum eigenstates

We start from $[P^\mu, P^\nu] = 0$, and diagonalize:

$$P^\mu |p^\mu\rangle = p^\mu |p^\mu\rangle.$$

We know how these states transform under translations, but we need to know how they transform under Lorentz.

4.2 Step 2: Casimir operator P^2

$P^2 \equiv (P^0)^2 - |\vec{P}|^2$ is a Casimir operator of the Poincaré group. We have $[P^2, P^\mu] = 0$.

We diagonalize together P^2, P^μ :

$$\begin{cases} P^\mu |p^2, p^\mu\rangle = p^\mu |p^2, p^\mu\rangle \\ P^2 |p^2, p^\mu\rangle = p^2 |p^2, p^\mu\rangle \end{cases} \quad p^2 = p_\mu p^\mu = (p^0)^2 - |\vec{p}|^2.$$

The value of p^2 is a Lorentz scalar (we do not change it by applying a Lorentz transformation). Therefore, it makes sense to introduce a first classification based on its value.

Added explanation (not in the original notes)

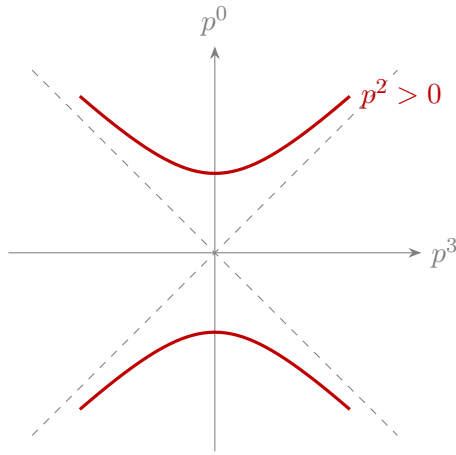
That P^2 is a Casimir requires it to commute with the $J^{\mu\nu}$ too, not only with the P^μ . Check with the bracket just derived: $[J^{\rho\sigma}, P^2] = [J^{\rho\sigma}, P_\mu P^\mu] = P_\mu [J^{\rho\sigma}, P^\mu] + [J^{\rho\sigma}, P_\mu] P^\mu = 2i(P^\rho P^\sigma - P^\sigma P^\rho) = 0$ (using $[P^\mu, P^\nu] = 0$). This is the Poincaré analogue of $|\vec{P}|^2$ commuting with J in $SE(2)$ (Lecture 1, p. 15), and by the same Schur-lemma logic its eigenvalue p^2 is one of the labels of an irrep. (The Poincaré group actually has a *second* Casimir, built from the Pauli–Lubanski vector, which will encode the spin — it is foreshadowed by the little-group theorem below.)

[p. 2*] There are four possible cases:

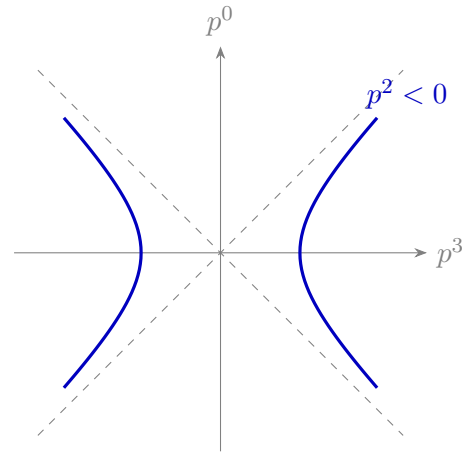
1. $p^2 = 0$ with $p^0 = |\vec{p}| = 0$ null case;
2. $p^2 > 0$ time-like case;
3. $p^2 = 0, p^0 \neq 0$ light-like case;
4. $p^2 < 0$ space-like case.

We can make a further classification based on the fact that, if $p^2 \geq 0 \Rightarrow \text{sign}(p^0)$ is Lorentz invariant.

Proof. Consider $p^2 = (p^0)^2 - |\vec{p}|^2$ with $p^1 = p^2 = 0$ (components), so that we have $p^2 = (p^0)^2 - (p^3)^2$, and write contours of constant p^2 in the p^0 – p^3 plane.



time-like hyperbolas:
two disconnected branches



space-like hyperbolas:
each branch crosses $p^0 = 0$

■ **Case $p^2 > 0$.** For fixed p^2 , we move along a branch with a Lorentz transformation (that alters p^0 and p^3 , keeping p^2 unchanged). Take, for instance, $p^0 > 0$ (we fix the sign of p^0) and select the upper branch. Lorentz transformations in $SO^\uparrow(1, 3)$ are [p. 3*] continuously connected to the identity, and cannot jump from a branch to the other.

■ **Case $p^2 < 0$.** In this case we sit on one of the blue branches, and it is totally possible to change $\text{sign}(p^0)$ with an $SO^\uparrow(1, 3)$ Lorentz transformation.

Added explanation (not in the original notes)

The topological point: for $p^2 > 0$ the locus of constant p^2 consists of *two* disconnected hyperbola branches, separated by the region around $p^0 = 0$ (a time-like vector can never have $p^0 = 0$, since that would force $p^2 \leq 0$). A continuous curve of Lorentz transformations starting at $\mathbb{I}$ drags p^μ along a continuous path on the locus, so it can never cross to the other branch: $\text{sign}(p^0)$ is an invariant label. For $p^2 < 0$ each branch passes through $p^0 = 0$, so the sign of p^0 can be flipped continuously — no extra label. Physically, this is why “the sign of the energy” is meaningful for massive particles. Note the restriction to $SO^\uparrow(1, 3)$ matters: the time-reversal component of the full Lorentz group *does* exchange the two branches — that is precisely what “non-orthochronous” means (Lecture 2, p. 34 box).

We focus on three cases:

timelike p	$(p^2 \equiv M^2 > 0, p^0 > 0)$	“massive case”
lightlike p	$(p^2 = 0, p^0 > 0)$	“massless case”
null p	$(p^0 = \vec{p} = 0)$	“the vacuum”

From now on, the discussion is specified for the massive case.

Added explanation (not in the original notes)

The space-like case $p^2 < 0$ is quietly dropped: it would describe “tachyons” (faster-than-light quanta), which are not realized as physical particles — and the sign-of-energy instability just proved (no invariant $\text{sign}(p^0)$) is one of the reasons they are problematic. The “vacuum” case $p^\mu = 0$ is invariant under the whole Lorentz group and carries the trivial one-dimensional representation: the empty state looks the same to every inertial observer.

4.3 Step 3: choice of a reference vector

[p. 3*]

We setup our notation. We write

$$\begin{cases} P^\mu |M, \vec{p}\rangle = p^\mu |M, \vec{p}\rangle, & M > 0 \\ P^2 |M, \vec{p}\rangle = M^2 |M, \vec{p}\rangle \end{cases}$$

Since $p^2 = M^2$, we have $(p^0)^2 - |\vec{p}|^2 = M^2$ and

$$p^0 = + (M^2 + |\vec{p}|^2)^{1/2}$$

[p. 4*] Consequently, given M and $\vec{p}$ is enough to reconstruct the full p^μ .

We now focus on a specific choice of reference momentum, p_{ref}^μ , that verifies $p^2 = M^2$ with $p^0 > 0$. In the massive case, an obvious choice is

$$p_{\text{ref}}^\mu = (M, \vec{0}), \quad p_{\text{ref}}^2 = M^2 \quad (\text{"}M\text{" identified with the rest mass}).$$

Specified for p_{ref} , the eigensystem is

$$P^\mu |M, \vec{0}\rangle = p_{\text{ref}}^\mu |M, \vec{0}\rangle, \quad P^2 |M, \vec{0}\rangle = M^2 |M, \vec{0}\rangle.$$

Added explanation (not in the original notes)

This is the third appearance of the same move: circle of radius p and $p_{\text{ref}} = (p, 0)$ for $SE(2)$ (Lecture 1, p. 19); now mass shell $(p^0)^2 - |\vec{p}|^2 = M^2$ (upper branch) and rest-frame momentum $(M, \vec{0})$ for Poincaré — exactly the dictionary anticipated in the Lecture 2 pivot box. The state $|M, \vec{0}\rangle$ is “the particle at rest”, and the boost of Lecture 2 (p. 43), $\Lambda_B(\eta, \hat{p})(M, \vec{0})^T = (E, \vec{p})^T$, is precisely the tool that will generate all the other momenta from it.

4.4 Step 4: identification of the “Wigner little group”

We look for Lorentz transformations that leave the reference vector invariant.

Definition. Given a four-vector p^μ , we define the *little group* $G(p)$ as the set of Lorentz transformations that leave p^μ invariant:

$$G(p) = \left\{ \Lambda \in SO^\uparrow(1, 3) : \Lambda^\mu{}_\nu p^\nu = p^\mu \right\}.$$

[p. 5*] (It is possible to show that $G(p)$ is a subgroup of the Lorentz group.) In the massive case, we see that

$$G(p_{\text{ref}}^\mu) = SO(3) \quad \text{massive case,}$$

since $p_{\text{ref}}^\mu = (M, \vec{0})$ is left invariant by spatial rotations.

Added explanation (not in the original notes)

Why the little group is the key idea. $G(p)$ is what mathematicians call the *stabilizer* (or isotropy group) of p . The logic of Wigner’s method of induced representations: the group action splits momentum space into *orbits* (here: the mass shells), on each orbit all points are equivalent (connected by boosts), so the only genuinely new freedom lives in the internal space at one point — and that internal space carries a representation of the little group. For

$SE(2)$ the little group of p_{ref} was trivial and the states carried no extra label; here it is $SO(3)$, and its irreps are labeled by *spin* — this is where the spin of a massive particle comes from. Subgroup check, one line: if $\Lambda_1 p = p$ and $\Lambda_2 p = p$ then $\Lambda_1 \Lambda_2 p = p$ and $\Lambda_1^{-1} p = p$; closure, identity and inverses done.

In view of future applications, let us state a theorem of general validity.

Theorem (algebra of the little group). The generators of the little group are the three components of the vector

$$\vec{W} \equiv \vec{J} + \frac{\vec{K} \times \vec{p}}{p^0}.$$

Proof. LG transformations are defined by

$$\Lambda^\mu{}_\nu p^\nu = p^\mu$$

(generic Lorentz transformation identified by 6 parameters) $\Rightarrow$ these equations provide constraints on the possible form of Λ . [p. 6*] Since we work at the level of the algebra, it is enough to consider infinitesimal transformations:

$$\Lambda^\mu{}_\nu p^\nu = (\delta^\mu{}_\nu + \omega^\mu{}_\nu) p^\nu = p^\mu + \omega^\mu{}_\nu p^\nu \stackrel{!}{=} p^\mu \quad \rightsquigarrow \quad \boxed{\omega^\mu{}_\nu p^\nu = 0}$$

Consequently, a little group transformation is generated by a linear combination of generators

$$\frac{1}{2} \omega_{\mu\nu} J^{\mu\nu} = d\vec{\theta} \cdot \vec{J}_{4\text{vec}} + d\vec{\eta} \cdot \vec{K}_{4\text{vec}}, \quad d\eta^i = \omega^{i0}, \quad d\theta^k = \frac{1}{2} \varepsilon_{kij} \omega^{ij} \quad (\star)$$

subject to $\omega^\mu{}_\nu p^\nu = 0$.

■ We analyze the constraints.

$$\mu = 0 \rightsquigarrow \omega^0{}_\nu p^\nu = \omega^0{}_i p^i = -\omega^{0i} p^i = +\omega^{i0} p^i = d\eta^i p^i = d\vec{\eta} \cdot \vec{p} = 0 \quad \rightsquigarrow \quad \boxed{d\vec{\eta} \cdot \vec{p} = 0}$$

$$\begin{aligned} \mu = i \rightsquigarrow \omega^i{}_\nu p^\nu &= \omega^i{}_0 p^0 + \omega^i{}_j p^j = \omega^{i0} p^0 - \omega^{ij} p^j = d\eta^i p^0 - \varepsilon_{kij} d\theta^k p^j \\ &= d\eta^i p^0 + \varepsilon_{ikj} d\theta^k p^j = d\eta^i p^0 + (d\vec{\theta} \times \vec{p})^i = 0. \end{aligned}$$

[p. 7*] In vector form,

$$d\vec{\eta} p^0 + d\vec{\theta} \times \vec{p} = 0 \quad \rightsquigarrow \quad \boxed{d\vec{\eta} = -\frac{d\vec{\theta} \times \vec{p}}{p^0}}$$

Out of the six parameters that define a Lorentz transformation, only 3 are independent. (Note that $d\vec{\eta} \cdot \vec{p} = 0$ is redundant once we fix $d\vec{\eta} = -d\vec{\theta} \times \vec{p}/p^0$.)

Back to ($\star$):

$$\begin{aligned} \frac{1}{2} \omega_{\mu\nu} J^{\mu\nu} &= d\vec{\theta} \cdot \vec{J}_{4\text{vec}} - \frac{(d\vec{\theta} \times \vec{p}) \cdot \vec{K}_{4\text{vec}}}{p^0} && \text{using circular shift of the triple product,} \\ &= d\vec{\theta} \cdot \vec{J}_{4\text{vec}} - \frac{d\vec{\theta} \cdot (\vec{p} \times \vec{K}_{4\text{vec}})}{p^0} && \vec{a} \cdot (\vec{b} \times \vec{c}) = \vec{c} \cdot (\vec{a} \times \vec{b}) = \vec{b} \cdot (\vec{c} \times \vec{a}) \\ &= d\vec{\theta} \cdot \left(\vec{J}_{4\text{vec}} + \frac{\vec{K}_{4\text{vec}} \times \vec{p}}{p^0} \right). \end{aligned}$$

This concludes the proof: a generic element of the little group can be written as a linear combination of the three generators

$$\vec{W} \equiv \vec{J} + \frac{\vec{K} \times \vec{p}}{p^0}$$

$\rightsquigarrow$ in the massive case ($\vec{p} = \vec{0}$): $\vec{W} = \vec{J}$, rotations.

Added explanation (not in the original notes)

Sanity checks and outlook. (i) At $\vec{p} = \vec{0}$ the theorem reproduces $G(p_{\text{ref}}) = SO(3)$ found directly: $\vec{W} = \vec{J}$. (ii) Counting: the constraint $\omega^\mu{}_\nu p^\nu = 0$ looks like 4 equations, but as the notes observe one of them ($\mu = 0$) is implied by the other three — consistency with the general fact that $\omega_{\mu\nu} p^\nu p^\mu = 0$ automatically (antisymmetric tensor contracted with the symmetric $p^\nu p^\mu$), so at most 3 constraints are independent, leaving $6 - 3 = 3$ generators for *any* p . (iii) The vector $\vec{W}$ is closely related to the *Pauli–Lubanski* four-vector $W^\mu \equiv -\frac{1}{2}\epsilon^{\mu\nu\rho\sigma} J_{\nu\rho} P_\sigma$: on momentum- p states, the spatial components of W^μ/p^0 reduce exactly to the $\vec{W}$ of the theorem, and $W_\mu W^\mu$ is the second Casimir of the Poincaré group, with eigenvalue $-M^2 j(j+1)$ — the covariant way of saying “the spin is j ”.

4.5 Diagonalizing the little group: spin

[p. 8*]

Back to the massive case: what is the utility of the little group? We start from the following observation. In general, $[\vec{J}, \vec{P}] \neq 0$, and they cannot be diagonalized together. $\hookrightarrow$ However, if we restrict to the subspace that corresponds to the reference vector, we have $\vec{p} = \vec{0}$ and it is possible to make a common diagonalization.

Since the $\vec{J}$ close the algebra of the angular momentum, we follow QM and diagonalize together $|\vec{J}|^2$ and J^3 :

$$\begin{aligned} P^\mu |M, \vec{0}, j, m\rangle &= p_{\text{ref}}^\mu |M, \vec{0}, j, m\rangle \\ P^2 |M, \vec{0}, j, m\rangle &= M^2 |M, \vec{0}, j, m\rangle \\ |\vec{J}|^2 |M, \vec{0}, j, m\rangle &= j(j+1) |M, \vec{0}, j, m\rangle \quad j = 0, \frac{1}{2}, 1, \dots \\ J^3 |M, \vec{0}, j, m\rangle &= m |M, \vec{0}, j, m\rangle \quad -j \leq m \leq +j \end{aligned}$$

Formally,

$$|M, \vec{0}, j, m\rangle = |M, \vec{0}\rangle \otimes |j, m\rangle.$$

Comment. We would like to know, eventually, how the states $|M, \vec{0}\rangle$ transform under Lorentz. We start here from a... (*the sentence continues in the next lecture*).

Added explanation (not in the original notes)

What has been achieved, and what the labels mean. On the rest-frame subspace the commuting set $\{P^\mu, P^2, |\vec{J}|^2, J^3\}$ is simultaneously diagonalized, and the state of a massive particle at rest is labeled by *mass* M (first Casimir), *spin* j (little-group Casimir — the second Poincaré Casimir in disguise), and the spin projection m . Note that half-integer j appears in the list $j = 0, \frac{1}{2}, 1, \dots$: strictly, the irreps of $SO(3)$ only allow integer j ; half-integer values belong to its double cover $SU(2)$, which is the right object to represent in quantum mechanics because states are defined up to a phase (projective representations). This is how electrons ($j = \frac{1}{2}$) fit into Wigner’s scheme.

The remaining task — announced by the final “Comment” — is the last step of the induced-representation construction, the Poincaré analogue of $|\vec{p}\rangle \equiv \mathcal{U}(\theta) |\vec{p}_{\text{ref}}\rangle$ from Lectures 1–2: *define* the moving states by applying the standard boost, $|M, \vec{p}, j, m\rangle \propto U(\Lambda_B) |M, \vec{0}, j, m\rangle$ (with the standard boost of Lecture 2, p. 43, possibly composed with the rotation $\hat{z} \rightarrow \hat{p}$ of p. 41), and then work out how a generic Lorentz transformation acts on them — producing a momentum jump plus a little-group (Wigner) rotation acting on the m index. That is presumably where the next lecture picks up.

(End of Lecture 3 — the notes continue with the transformation of the states $|M, \vec{0}\rangle$ under Lorentz.)